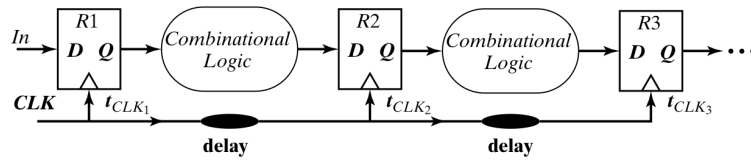
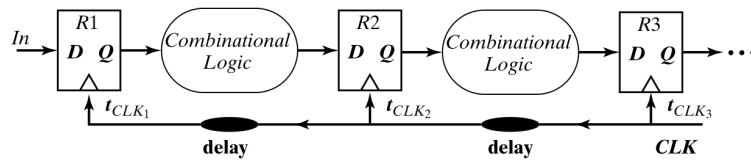


Positive and Negative Skew



(a) Positive skew



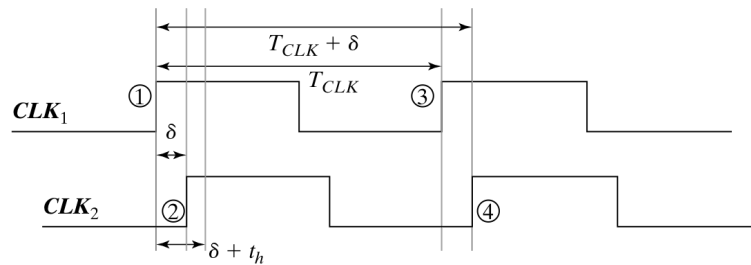
(b) Negative skew

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Positive Skew



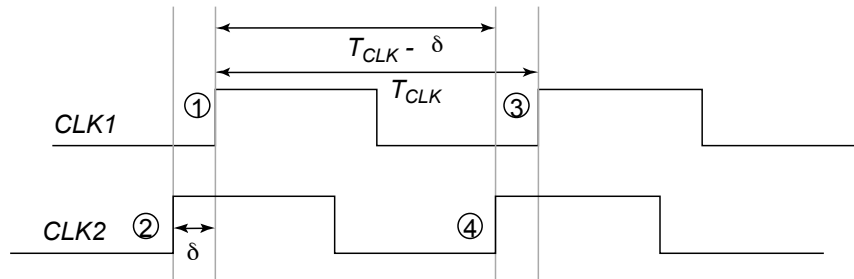
Launching edge arrives before the receiving edge

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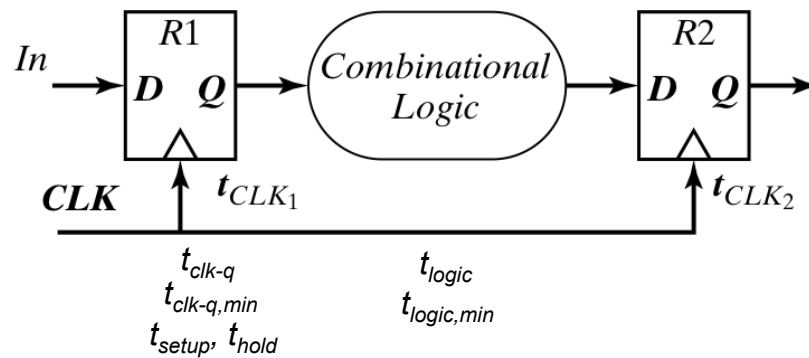
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Negative Skew



Receiving edge arrives before the launching edge

Timing Constraints

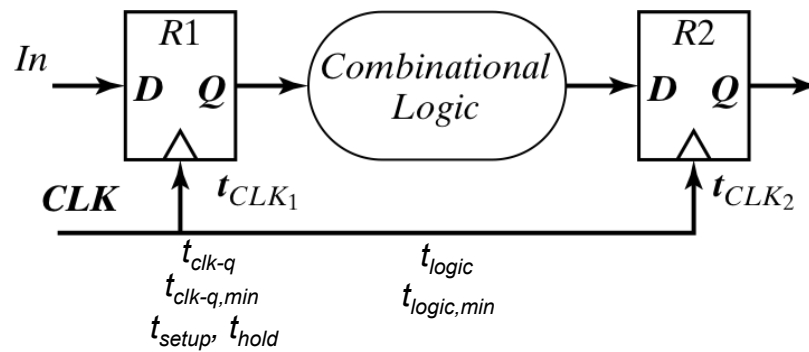


Minimum cycle time:

$$T_{clk} + \delta = t_{clk-q} + t_{setup} + t_{logic}$$

Worst case is when receiving edge arrives early (negative δ)

Timing Constraints



Hold time constraint:

$$t_{(clk-q,min)} + t_{(logic,min)} > t_{hold} + \delta$$

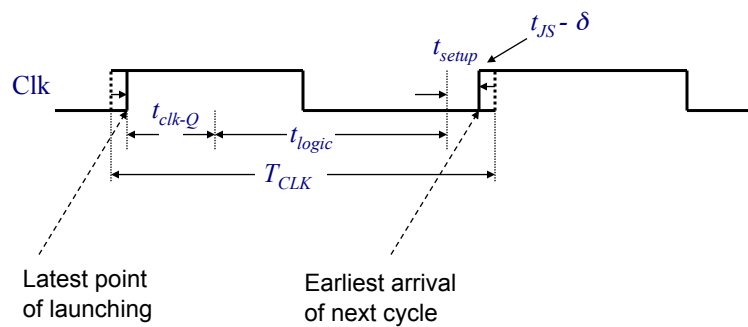
Worst case is when receiving edge arrives late
Race between data and clock

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Longest Logic Path in Edge-Triggered Systems



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Clock Constraints in Edge-Triggered Systems

If launching edge is late and receiving edge is early, the data will not be too late if:

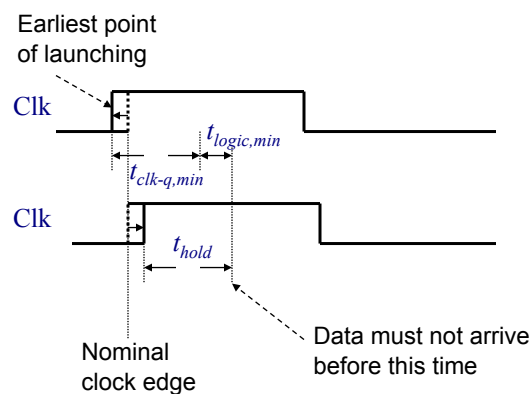
$$t_{clk-q} + t_{logic} + t_{setup} < T_{CLK} - t_{JS,1} - t_{JS,2} + \delta$$

Minimum cycle time is determined by the maximum delays through the logic

$$t_{clk-q} + t_{logic} + t_{setup} - \delta + 2t_{JS} < T_{CLK}$$

Skew can be either positive or negative

Shortest Path



Clock Constraints in Edge-Triggered Systems

If launching edge is early and receiving edge is late:

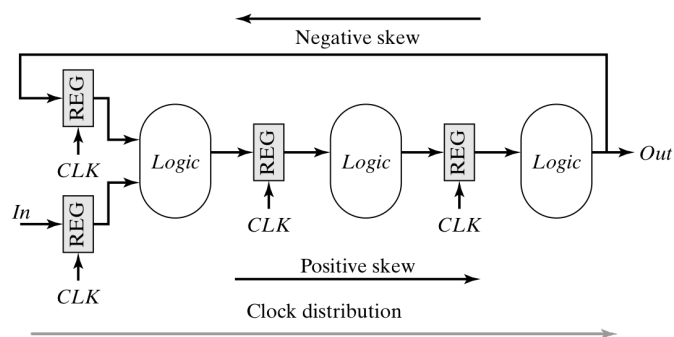
$$t_{clk-q,min} + t_{logic,min} - t_{JS,1} > t_{hold} + t_{JS,2} + \delta$$

Minimum logic delay

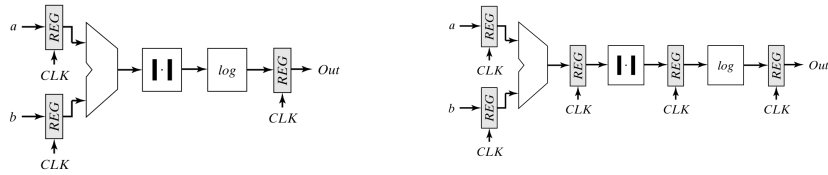
$$t_{clk-q,min} + t_{logic,min} > t_{hold} + 2t_{JS} + \delta$$

(This assumes jitter at launching and receiving clocks are independent – which usually is not true)

Datapath with Feedback



Pipelining



Reference

Clock Period	Adder	Absolute Value	Logarithm
1	$a_1 + b_1$		
2	$a_2 + b_2$	$ a_1 + b_1 $	
3	$a_3 + b_3$	$ a_2 + b_2 $	$\log(a_1 + b_1)$
4	$a_4 + b_4$	$ a_3 + b_3 $	$\log(a_2 + b_2)$
5	$a_5 + b_5$	$ a_4 + b_4 $	$\log(a_3 + b_3)$

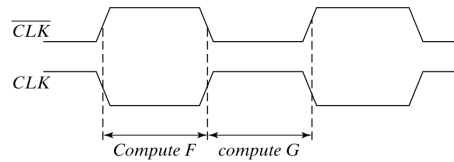
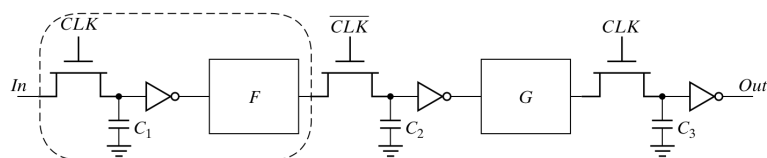
Pipelined

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Latch-Based Clocking



(Domino logic almost always uses latch-based clocking)

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Latch vs. Flip-flop

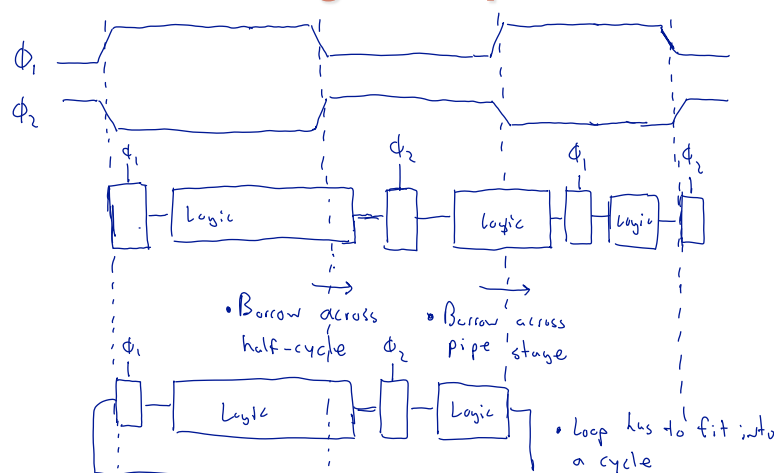
- In a flip-flop based system:
 - Data launches on one rising edge
 - And must arrive before next rising edge
 - If data arrives late, system fails
 - If it arrives early, wasting time
 - Flip-flops have hard edges
- In a latch-based system:
 - Data can pass through latch while it is transparent
 - Long cycle of logic can borrow time into next cycle
 - As long as each loop finished in one cycle

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Time Borrowing Example



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Latch vs. Flip-flop Summary

- ❑ Flip-flops generally easier to use
 - Most digital ASICs designed with register-based timing
- ❑ But, latches (both pulsed and level-sensitive) allow more flexibility
 - And hence can potentially achieve higher performance
 - Latches can also be made more tolerant of clock un-certainty
 - More in EE241

CMOS Transistor Scaling



Goals of Technology Scaling

- ❑ Make things cheaper:
 - Want to sell more functions (transistors) per chip for the same money
 - Or build same products cheaper
 - Price of a transistor has to be reduced
- ❑ But also want to be faster, smaller, lower power...

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Technology Scaling

- ❑ Benefits of 30% “Dennard” scaling (1974):
 - Double transistor density
 - Reduce gate delay by 30% (increase operating frequency by 43%)
 - Reduce energy per transition by 65% (50% power savings @ 43% increase in frequency)
- ❑ Die size used to increase by 14% per generation (not any more)
- ❑ Technology generation spans 2-3 years

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Moore was not always accurate

Projected 2000 Wafer, circa 1975

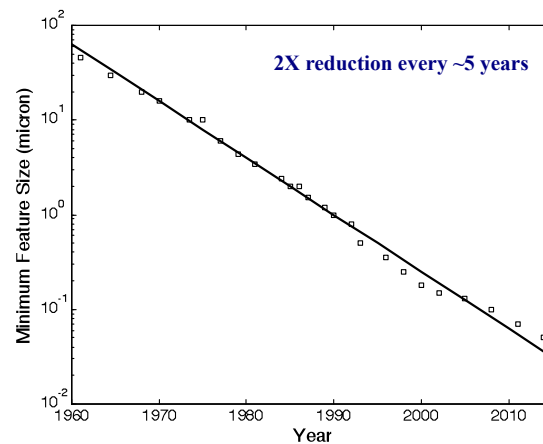


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Technology Scaling (1)

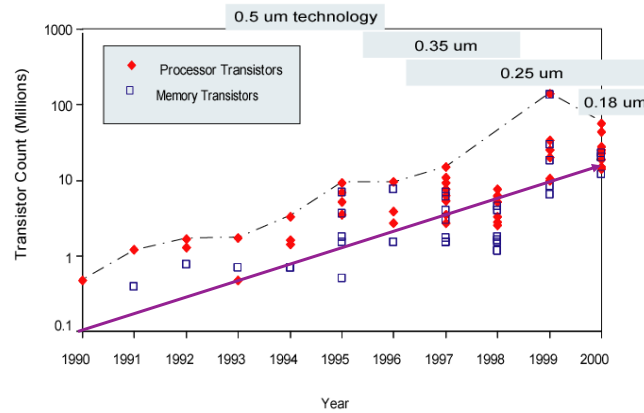
**Minimum Feature Size**

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Technology Scaling (2)



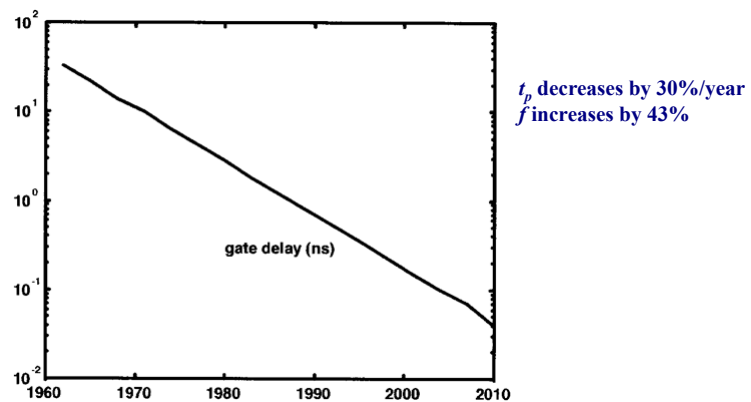
Number of components per chip

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Technology Scaling (3)



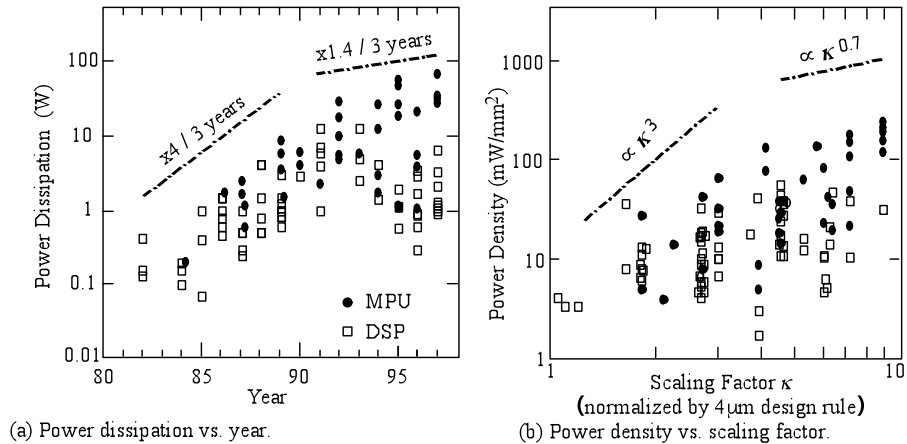
Propagation Delay

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Technology Scaling (4)



From Kuroda

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Technology Scaling Models

- **Full Scaling (Constant Electrical Field)**

ideal model — dimensions and voltages scale together by the same factor S

- **Fixed Voltage Scaling**

most common model until 1990's
only dimensions scale, voltages remain constant

- **General Scaling**

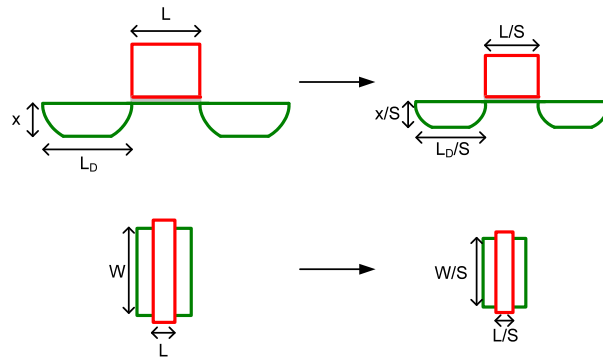
most realistic for today's situation —
voltages and dimensions scale with different factors

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Scaling



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Full Scaling (Dennard, Long-Channel)

- W, L, t_{ox} : $1/S$
- V_{DD}, V_T : $1/S$
- Area: WL
- C_{ox} : $1/tox$
- C_L : $C_{ox}WL$
- I_D : $C_{ox}(W/L)(V_{DD}-V_T)^2$
- R_{eq} : V_{DD}/I_{DSAT}

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Full Scaling (Dennard, Long-Channel)

- $W, L, t_{ox}: 1/S$
- $V_{DD}, V_T: 1/S$
- $t_p: R_{eq} C_L$
- $P_{avg}: C_L V_{DD}^2 / t_p$
- $P_{avg}/A: C_{ox} V_{DD}^2 / t_p$

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Scaling Relationships for Long Channel Devices

Parameter	Relation	Full Scaling	General Scaling	Fixed Voltage Scaling
W, L, t_{ox}		$1/S$	$1/S$	$1/S$
V_{DD}, V_T		$1/S$	$1/U$	1
N_{SUB}	V/W_{depl}^2	S	S^2/U	S^2
Area/Device	WL	$1/S^2$	$1/S^2$	$1/S^2$
C_{ox}	$1/t_{ox}$	S	S	S
C_L	$C_{ox}WL$	$1/S$	$1/S$	$1/S$
k_n, k_p	$C_{ox}W/L$	S	S	S
I_{av}	$k_{n,p} V^2$	$1/S$	S/U^2	S
t_p (intrinsic)	$C_L V / I_{av}$	$1/S$	U/S^2	$1/S^2$
P_{av}	$C_L V^2 / t_p$	$1/S^2$	S/U^3	S
PDP	$C_L V^2$	$1/S^3$	$1/SU^2$	$1/S$

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Full Scaling (Dennard, Short-Channel)

- $W, L, t_{ox}: 1/S$
- $V_{DD}, V_T: 1/S$
- Area: WL
- $C_{ox}: 1/t_{ox}$
- $C_L: C_{ox}WL$
- $I_D: WC_{ox}v_{sat}(V_{DD}-V_T-V_{SAT}/2)$
- $R_{eq}: V_{DD}/I_{DSAT}$

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Full Scaling (Dennard, Short-Channel)

- $W, L, t_{ox}: 1/S$
- $V_{DD}, V_T: 1/S$
- $t_p: R_{eq}C_L$
- $P_{avg}: C_L V_{DD}^2/t_p$
- $P_{avg}/A: C_{ox} V_{DD}^2/t_p$

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Transistor Scaling (Velocity-Saturated Devices)

Parameter	Relation	Full Scaling	General Scaling	Fixed-Voltage Scaling
W, L, t_{ox}		$1/S$	$1/S$	$1/S$
V_{DD}, V_T		$1/S$	$1/U$	1
N_{SUB}	V/W_{dop}^2	S	S^2/U	S^2
Area/Device	WL	$1/S^2$	$1/S^2$	$1/S^2$
C_{ox}	$1/t_{ox}$	S	S	S
C_{gate}	$C_{ox}WL$	$1/S$	$1/S$	$1/S$
k_n, k_p	$C_{ox}W/L$	S	S	S
I_{sat}	$C_{ox}WV$	$1/S$	$1/U$	1
Current Density	$I_{sat}/Area$	S	S^2/U	S^2
R_{on}	V/I_{sat}	1	1	1
Intrinsic Delay	$R_{on}C_{gate}$	$1/S$	$1/S$	$1/S$
P	$I_{sat}V$	$1/S^2$	$1/U^2$	1
Power Density	$P/Area$	1	S^2/U^2	S^2

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An interesting question

- ❑ What ~~will~~ did cause this model to break?
 - Leakage set by kT/q
 - Temp. does not scale
 - V_T set to minimize power
 - Power actually increased
 - Leakage increased drastically
 - f increased faster than device speed
 - Hit cooling limit
 - Process Variation
 - Hard to build very small things accurately (less averaging)

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